

2(a)(i)

gametes correctly entered ;

genotypes correct ; **A** ecf

phenotypes match genotypes ; **A** ecf

ratio of fruit colours: 12 white : 3 yellow : 1 green ; **A** ecf

	AB	Ab	aB	ab
AB	AABB white	AABb white	AaBB white	AaBb white
Ab	AABb white	AAbb white	AaBb white	Aabb white
aB	AaBB white	AaBb white	aaBB yellow	aaBb yellow
ab	AaBb white	Aabb white	aaBb yellow	aabb green

Question	Answer	Marks
2(a)(ii)	plant P: AaBB ; plant Q: Aabb ;	2
2(a)(iii)	random / independent, assortment ; metaphase I ;	2
2(b)(i)	<i>Freedom II because:</i> 1 contains, viral / foreign, DNA / genes <i>or</i> has DNA from 2 different, sources / organisms ; 2 <i>ref. to</i> genetic engineering / GMO / genetic modification / transgenic ;	2
2(b)(ii)	<i>advantage:</i> improved yield / plants do not die of disease / crop not lost ; <i>disadvantage:</i> viral proteins might cause allergies / may be social resistance to GM crops ;	2

Question	Answer	Marks
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